

SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Precision (Level 3)
Question Count: 50

Total Marks: 100

Code of Conduct

I **JAIVANT.S (192311326)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **JAIVANT.S**

Assessment Tasks

MCQ Test

1. A company wants to run several virtual machines on a single physical server to reduce hardware costs. Which technology should it use?

- A) Encryption
- B) Virtualization
- C) Authentication
- D) Compression

ANSWER: B) Virtualization

Justification: Virtualization allows multiple virtual machines to run on one physical server, improving resource utilization and reducing hardware costs.

2. A student wants to access a cloud-based document application from a laptop without installing the application. Which component is most suitable?

- A) Web browser
- B) CPU
- C) RAM
- D) Storage array

ANSWER: A) Web browser

Justification: A web browser allows the student to access a cloud application through the Internet without installing the application locally.

3. A hospital stores patient records in the cloud and wants to prevent unauthorized users from reading the data. Which technique should be used?

- A) Encryption
- B) Load balancing
- C) Caching
- D) Compression

ANSWER: A) Encryption

Justification: Encryption converts data into an unreadable form so that unauthorized users cannot understand the information.

4. An organization experiences heavy traffic on its cloud application and wants to distribute requests across several servers. Which technology should it implement?

- A) Firewall
- B) Load balancer
- C) Web browser
- D) Storage array

ANSWER: B) Load balancer

Justification: A load balancer distributes incoming requests across multiple servers, preventing any single server from becoming overloaded.

5. A startup wants to rent virtual machines, storage, and networking resources instead of purchasing physical infrastructure. Which cloud service model is appropriate?

- A) SaaS
- B) PaaS
- C) IaaS
- D) API

ANSWER: C) IaaS

Justification: Infrastructure as a Service provides virtual machines, storage, networking, and other infrastructure resources on demand.

6. A user wants to access an online cloud storage service from a desktop computer. What does the user primarily require?

- A) Web browser and Internet connection
- B) Printer and scanner
- C) BIOS and CPU
- D) Keyboard and mouse only

ANSWER: A) Web browser and Internet connection

Justification: Cloud storage is accessed over the Internet, commonly through a browser or cloud application.

7. A software company wants an environment containing development tools and runtime services to build and deploy applications. Which model should it select?

- A) IaaS
- B) PaaS
- C) SaaS
- D) SAN

ANSWER: B) PaaS

Justification: PaaS provides developers with development tools and a runtime environment without requiring them to manage the underlying infrastructure.

8. A teacher uses an online application to create, edit, and share documents with students without installing desktop software. What type of application is being used?

- A) Web application
- B) Device driver
- C) Operating system
- D) Firmware

ANSWER: A) Web application

Justification: A web application runs through a web browser and does not require traditional desktop installation.

9. An e-commerce application needs to obtain payment information from an external payment service. Which technology can facilitate this communication?

- A) Web API
- B) RAM
- C) BIOS
- D) Monitor

ANSWER: A) Web API

Justification: A Web API allows an application to communicate and exchange data with an external service.

10. A user enters a cloud application's web address into Chrome and accesses the application. What role does Chrome perform?

- A) Cloud server
- B) Cloud client
- C) Storage array
- D) Database server

ANSWER: B) Cloud client

Justification: Chrome acts as the client by sending requests to the cloud application and displaying the responses to the user.

11. A company wants employees to access files from different offices without maintaining local storage servers at every office. Which solution is most appropriate?

- A) Cloud storage
- B) Local RAM
- C) BIOS storage
- D) CPU cache

ANSWER: A) Cloud storage

Justification: Cloud storage allows employees in different locations to access centrally stored files through the Internet.

12. A bank requires high-speed block-level storage that can be shared by multiple servers. Which technology should it use?

- A) SAN
- B) Web browser
- C) SaaS
- D) HTML

ANSWER: A) SAN

Justification: A Storage Area Network provides high-speed block-level storage that can be accessed by multiple servers.

13. A data center is designing a SAN and needs a centralized device that provides storage capacity to connected servers. Which component is required?

- A) Storage array
- B) Web browser
- C) Keyboard
- D) Application server only

ANSWER: A) Storage array

Justification: A storage array provides and manages the storage capacity used by connected servers in a SAN environment.

14. Five students located in different cities need to work simultaneously on the same project document. Which cloud feature would help them most?

- A) Cloud collaboration
- B) Local-only storage
- C) BIOS configuration
- D) CPU scheduling

ANSWER: A) Cloud collaboration

Justification: Cloud collaboration allows users in different locations to work on shared documents simultaneously.

15. An organization wants applications from different cloud providers to communicate effectively. What should it emphasize?

- A) Cloud standards
- B) Hardware replacement
- C) Screen resolution
- D) Local storage

ANSWER: A) Cloud standards

Justification: Cloud standards improve compatibility and interoperability between different cloud services and providers.

16. A business experiences unpredictable demand during festival seasons. It wants resources to increase during peak periods and decrease afterward. Which cloud characteristic is most useful?

- A) Elasticity
- B) Manual processing
- C) Local storage
- D) Fixed capacity

ANSWER: A) Elasticity

Justification: Elasticity allows resources to expand when demand increases and reduce when demand decreases.

17. A company subscribes to an online email application and does not manage the underlying servers or operating system. Which model is being used?

- A) IaaS
- B) PaaS
- C) SaaS
- D) SAN

ANSWER: C) SaaS

Justification: SaaS provides complete software applications through the Internet while the provider manages the infrastructure and platform.

18. A developer wants to deploy an application without managing physical servers or operating systems. Which cloud model is most suitable?

- A) PaaS
- B) SAN
- C) LAN
- D) DNS

ANSWER: A) PaaS

Justification: PaaS allows developers to deploy applications without managing the underlying physical infrastructure and operating system.

19. A company wants only authorized employees to access its cloud application. Which security mechanism should be implemented?

- A) Authentication
- B) Compression
- C) Caching
- D) Load balancing

ANSWER: A) Authentication

Justification: Authentication verifies the identity of users before allowing them to access the application.

20. After logging into a cloud system, an employee should only be allowed to access financial reports. Which security process determines what the employee can access?

- A) Authorization
- B) Encryption
- C) Compression
- D) Routing

ANSWER: A) Authorization

Justification: Authorization determines which resources and actions an authenticated user is permitted to access.

21. A company wants to block unauthorized network traffic entering its cloud infrastructure. Which security component should it deploy?

- A) Firewall
- B) Web browser
- C) Storage array
- D) CPU

ANSWER: A) Firewall

Justification: A firewall monitors network traffic and blocks connections that do not meet defined security rules.

22. A bank's customers access their accounts through a website. The bank wants communication between customers and the server to be encrypted. Which protocol should be used?

- A) HTTP
- B) HTTPS
- C) FTP
- D) SMTP

ANSWER: B) HTTPS

Justification: HTTPS provides encrypted communication between the user's browser and the web server, making it suitable for banking information.

23. A mobile application retrieves weather information from an external cloud service. Which technology is most likely used?

- A) Web API
- B) RAM
- C) BIOS
- D) SAN

ANSWER: A) Web API

Justification: A Web API allows the mobile application to request weather information from the external cloud service.

24. An organization develops a mobile application that communicates with a cloud server using HTTP requests. What is being used?

- A) Web API
- B) Storage array
- C) CPU cache
- D) Virtual memory

ANSWER: A) Web API

Justification: Web APIs commonly use HTTP requests to enable communication between applications and cloud services.

25. A company wants employees to access business files from their office, home, and mobile devices. Which solution is most appropriate?

- A) Cloud storage
- B) Local hard disk only
- C) Offline storage
- D) BIOS

ANSWER: A) Cloud storage

Justification: Cloud storage allows users to access business files from different locations and devices through the Internet.

26. An online business expects its data volume to grow rapidly. Which cloud storage advantage would be most useful?

- A) Scalability
- B) Fixed storage capacity
- C) Manual backup only
- D) Local-only access

ANSWER: A) Scalability

Justification: Scalability allows storage capacity to increase as the business generates more data.

27. A video processing company needs very fast access to large amounts of data from multiple servers. Which solution is suitable?

- A) SAN
- B) Web browser
- C) SaaS
- D) Email

ANSWER: A) SAN

Justification: SAN provides high-speed storage access and is suitable for applications requiring large amounts of data to be accessed by multiple servers.

28. A project team wants members to edit a shared presentation and see updates immediately. Which capability should they use?

- A) Real-time cloud collaboration
- B) Local file transfer
- C) BIOS configuration
- D) Offline backup

ANSWER: A) Real-time cloud collaboration

Justification: Real-time cloud collaboration allows multiple users to edit shared files and see changes immediately.

29. A company wants to migrate its application between different cloud providers without major compatibility problems. What is important?

- A) Interoperability standards
- B) Screen resolution
- C) Keyboard type
- D) Local printer

ANSWER: A) Interoperability standards

Justification: Interoperability standards help applications and services work across different cloud environments with fewer compatibility problems.

30. A small company cannot afford to purchase expensive servers but needs computing resources for its new application. What is a major advantage of cloud computing for this company?

- A) Pay-as-you-use resource availability
- B) Mandatory hardware purchase
- C) Permanent fixed capacity
- D) No Internet access

ANSWER: A) Pay-as-you-use resource availability

Justification: Cloud computing allows the company to use computing resources without purchasing expensive physical infrastructure and pay according to usage.

31. A developer creates an application that automatically increases the number of instances when user traffic becomes high. Which cloud concept is being applied?

- A) Auto-scaling
- B) Encryption
- C) Authentication
- D) Data compression

ANSWER: A) Auto-scaling

Justification: Auto-scaling automatically increases or decreases the number of computing instances according to demand.

32. A development team wants its application to be scalable, resilient, and optimized for cloud infrastructure. Which approach should it follow?

- A) Cloud-native development
- B) Hardware-only development
- C) Offline development
- D) Manual deployment only

ANSWER: A) Cloud-native development

Justification: Cloud-native development focuses on applications designed specifically for cloud scalability, resilience, automation, and flexibility.

33. A large application is difficult to maintain because all functionality is contained in one large program. The development team wants to divide it into smaller independently deployable components. Which architecture should it choose?

- A) Microservices
- B) SAN
- C) Monolithic storage
- D) BIOS architecture

ANSWER: A) Microservices

Justification: Microservices divide a large application into smaller independently deployable services, making it easier to maintain and scale.

34. A government organization wants its cloud infrastructure to be used exclusively by its departments. Which deployment model is appropriate?

- A) Private cloud
- B) Public cloud
- C) Hybrid cloud
- D) Community website

ANSWER: A) Private cloud

Justification: A private cloud is dedicated to a single organization and provides greater control over access and infrastructure.

35. A company stores sensitive financial information in its private cloud but uses a public cloud for its website during high-traffic periods. Which model is being used?

- A) Private cloud
- B) Public cloud
- C) Hybrid cloud
- D) Community cloud

ANSWER: C) Hybrid cloud

Justification: A hybrid cloud combines private and public cloud environments, allowing sensitive data to remain private while public resources handle additional demand.

36. A company uses services from two different cloud providers for computing and backup. What type of cloud strategy is this?

- A) Private cloud
- B) Multi-cloud
- C) Single cloud
- D) Local computing

ANSWER: B) Multi-cloud

Justification: Multi-cloud means using services from two or more different cloud providers.

37. A user enters a URL into a browser, and the browser sends a request to a cloud server. What is the browser acting as?

- A) Client
- B) Database
- C) Storage array
- D) Firewall

ANSWER: A) Client

Justification: The browser acts as a client because it sends requests to the cloud server and receives responses.

38. An online shopping application must remain available even when one server fails. What approach should the company use?

- A) Multiple servers with redundancy
- B) One server only
- C) Local storage only
- D) Manual data entry

ANSWER: A) Multiple servers with redundancy

Justification: Redundant servers ensure that another server can continue providing the service if one server fails.

39. A researcher stores confidential research data in cloud storage and wants to protect it even if unauthorized people obtain the storage files. What should be implemented?

- A) Encryption
- B) Load balancing

- C) Auto-scaling
- D) Caching

ANSWER: A) Encryption

Justification: Encryption protects confidential data by making stored information unreadable to unauthorized users.

40. A company wants to detect whether cloud-stored files have been modified without authorization. Which mechanism is useful?

- A) Hashing
- B) Load balancing
- C) Virtualization
- D) Scaling

ANSWER: A) Hashing

Justification: Hashing generates a value based on file contents. A change to the file normally produces a different hash, helping detect modification.

41. A university wants students to use an online learning management application without installing or maintaining the software. Which service model is most appropriate?

- A) SaaS
- B) IaaS
- C) PaaS
- D) SAN

ANSWER: A) SaaS

Justification: SaaS provides a complete application online, while the service provider handles its maintenance and underlying infrastructure.

42. A developer has completed a cloud application and wants to identify errors before releasing it to users. What should be performed?

- A) Application testing
- B) Hardware replacement
- C) Data deletion
- D) Network disconnection

ANSWER: A) Application testing

Justification: Application testing helps identify errors, bugs, and performance issues before the application is released.

43. A company wants to recover important files after accidental deletion or system failure. Which service should it implement?

- A) Cloud backup
- B) Web browser

- C) Load balancing
- D) API gateway only

ANSWER: A) Cloud backup

Justification: Cloud backup keeps copies of important files so they can be restored after accidental deletion or system failure.

44. A cloud administrator wants to identify which virtual machines are consuming excessive CPU resources. Which activity is required?

- A) Resource monitoring
- B) Data encryption
- C) Web browsing
- D) File compression

ANSWER: A) Resource monitoring

Justification: Resource monitoring tracks CPU and other resource usage to identify virtual machines consuming excessive resources.

45. A company wants cloud resources to be automatically created whenever application demand increases. Which approach is most suitable?

- A) Cloud automation
- B) Manual configuration
- C) Local-only storage
- D) Offline processing

ANSWER: A) Cloud automation

Justification: Cloud automation can automatically create and configure resources when they are needed, reducing manual effort.

46. A company wants to use an online CRM system without purchasing servers or installing the application on employee computers. Which service is suitable?

- A) SaaS
- B) SAN
- C) IaaS only
- D) LAN

ANSWER: A) SaaS

Justification: SaaS provides complete software applications online, so the company does not need to install or maintain the application infrastructure.

47. A cloud application becomes slow because too many users are accessing a single server. Which solution would improve traffic distribution?

- A) Load balancing
- B) Data encryption

- C) Authentication
- D) File compression

ANSWER: A) Load balancing

Justification: Load balancing distributes incoming requests across multiple servers, improving performance and preventing a single server from becoming overloaded.

48. A financial application uses an API to exchange sensitive customer information. Which practice is most important?

- A) Use secure HTTPS communication and authentication
- B) Disable authentication
- C) Use unencrypted HTTP only
- D) Allow unrestricted access

ANSWER: A) Use secure HTTPS communication and authentication

Justification: HTTPS protects sensitive information during transmission, while authentication ensures that only authorized users or applications can access the API.

49. A development team wants to deploy an application quickly without manually configuring every server. Which cloud capability would help?

- A) Automated deployment
- B) Manual installation
- C) Local-only execution
- D) Hardware replacement

ANSWER: A) Automated deployment

Justification: Automated deployment reduces manual configuration and allows applications to be deployed quickly and consistently.

50. A company wants to develop a web application, store its data in the cloud, provide access through browsers, communicate with external services using APIs, and automatically scale during high demand. Which approach best satisfies these requirements?

- A) Cloud-based application architecture
- B) Standalone desktop architecture
- C) Local-only storage architecture
- D) Offline computing architecture

ANSWER: A) Cloud-based application architecture

Justification: A cloud-based architecture supports cloud storage, browser access, API communication, and automatic scaling, satisfying all the given requirements.

MCQ Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30%	Answers reflect strong and accurate concept mastery	Mostly accurate understanding	Partial understanding	Weak understanding
Accuracy of Responses	30%	Nearly all responses are correct	Most responses are correct	Some responses are correct	Few responses are correct
Application Awareness	15%	Shows strong recognition of practical cloud use cases	Reasonable recognition	Limited recognition	Weak recognition
Time Management	10%	Completes assessment efficiently	Completes on time	Needs extra time	Unable to finish well
Question Interpretation	15%	Interprets all items correctly	Misreads very few items	Misreads several items	Frequently misinterprets questions